

Mostafa Saad

AI Engineer & Data Scientist

01278447670 ◇ mostafasaadkh24@gmail.com ◇ Egypt ◇ [GitHub](#) ◇ [LinkedIn](#)

SUMMARY

Skilled in Data Science, Machine Learning, and Deep Learning. Experienced in Exploratory Data Analysis (EDA), predictive modeling, analytics, deployment, and AI in Cybersecurity, with a proven ability to deliver actionable insights and manage projects effectively.

EDUCATION

Bachelor in Computer Science, Higher Institute of Computers and Information (GPA: 3.2)

Sep '20 — Oct '24
Kafr El-Sheikh, Egypt

EXPERIENCE

AI Engineer

Digital Fortress

Jan '26 — Present
United Arab Emirates

- Architected and deployed phishing email detection solutions, advancing early threat identification and reinforcing enterprise cyber defense strategies
- Engineered end-to-end pipelines leveraging TF-IDF, encoder-based text features, Logistic Regression, ensemble classifiers, and transformer architectures for advanced representation learning
- Delivered up to 97% accuracy and a high F1-score in phishing classification, producing automated risk scores and threat labels to support security decision-making
- Integrated models into a production-ready backend via RESTful APIs and a functional demo application, enabling real-time phishing detection and seamless platform integration

NTI Scholarship — AI in Cybersecurity

National Telecommunication Institute

Sep '25 — Dec '25
Alexandria, Egypt

- Completed **288 hrs technical & 132 hrs professional** training applying AI/ML (CNNs, RNNs, GANs, transformers, diffusion models, LLM fine-tuning) to cybersecurity challenges.
- Hands-on in threat detection, cyber automation (Python, AI agents), adversarial AI, secure prompt engineering, RAG systems.
- Executed malware, phishing & social engineering detection; conducted capstone cyber operations projects.
- Strengthened professional skills: Business English, leadership, project management, career development, critical thinking.

SKILLS

Data Analysis & Visualization Data preprocessing, data cleaning, feature engineering, EDA, statistical analysis, Power BI

Machine Learning & Deep Learning Supervised/unsupervised learning, regression, classification, clustering, CNN, RNN, Transformers, hyperparameter tuning, ensemble methods (Random Forest, XGBoost, LightGBM, CatBoost)

AI in Cybersecurity Phishing detection, threat classification, risk scoring, malware analysis, penetration testing, real-time model deployment

Programming & Tools Python, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Flask, FastAPI, model evaluation, API development, web integration

PROJECTS

Phishing Email Detection [Link](#)

- Built an end-to-end detection pipeline using 40,000+ emails (2018-2025) from multiple sources
- Performed data cleaning, preprocessing, and EDA to ensure quality and balanced classes
- Developed two independent ML models with TF-IDF feature extraction, achieving 99% and 97% accuracy
- Implemented a Voting Ensemble to combine outputs and maximize prediction reliability. Validated the system through a functional demo application showcasing real-time phishing detection

Malware Detection System [Link](#)

- Developed an API to analyze Windows executable files (.exe) and classify them as Clean or Malware
- Combined static PE feature extraction with an ensemble machine learning model for improved detection accuracy
- Integrated threat intelligence APIs (VirusTotal, AlienVault OTX) to enrich analysis before ML inference

- Achieved ~99.5% accuracy with probability-based predictions for reliable malware classification
- Built using Python, FastAPI, and ML frameworks, enabling real-time deployment and enterprise integration

Human Personality Analysis & Classification [Link](#)

- Analyzed behavioral data using EDA, visualization, preprocessing, and feature engineering . Built and compared ML models (Logistic Regression, Random Forest, XGBoost, KNN) for personality type classification (Extrovert vs. Introvert)
- Achieved 92% accuracy and a best F1-score of 0.939 with KNN
- Identified emotional responses as strongest predictors, highlighting applications in psychology, HR, and marketing

CERTIFICATIONS

Data Analysis , Microsoft Egypt	Feb '25
AI in Cybersecurity , NTI Scholarship — AI in Cybersecurity	Dec '25

MILITARY STATUS

Permanently Exempt

LANGUAGES

English - Proficient